IRR based Filtering
(an intro to RPSL tools)
BGPQ4

BGP Deployment Workshop

UNIVERSITY OF OREGON

Last updated 18th August 2025

NSRC
Network Startup Resource Center

Manual Filters

- Verify the resource holder (WHOIS lookup) and craft filters

```
whois -h whois.apnic.net AS18024

aut-num:        AS18024
as-name:        BTTELECOM-AS-AP
descr:          Bhutan Telecom Ltd
country:        BT
org:            ORG-BTL2-AP
admin-c:        DNO1-AP
tech-c:         DNO1-AP
abuse-c:        AB1276-AP
mnt-lower:      MAINT-BT-DRUKNET
mnt-routes:     MAINT-BT-DRUKNET
mnt-by:         APNIC-HM
mnt-irt:        IRT-BTTELECOM-BT
last-modified:  2021-01-14T06:16:00Z
source:         APNIC
```

```
whois -h whois.apnic.net 202.144.128.0

inetnum:        202.144.128.0 - 202.144.129.255
netname:        DRUKNET
descr:          DrukNet System
descr:          DrukNet
descr:          Bhutan Telecom
descr:          Thimphu
country:        BT
admin-c:        JT106-AP
tech-c:         JT106-AP
abuse-c:        AB1276-AP
status:         ASSIGNED NON-PORTABLE
mnt-by:         MAINT-BT-DRUKNET
mnt-irt:        IRT-BTTELECOM-BT
last-modified:  2021-01-14T06:15:57Z
source:         APNIC
```

IRR (Internet Routing Registry) lookup

- Where networks publish routing intents
  - Route origination (route/route6 objects)

```
route:          202.144.128.0/20
descr:          DRUKNET-BLOCK-A1
country:        BT
notify:         ioc@bt.bt
mnt-by:         MAINT-BT-DRUKNET
origin:         AS18024
last-modified:  2018-09-18T09:37:40Z
source:         APNIC
```

```
route6:         2405:d000::/32
descr:          DRUKNET-IPV6-BLOCK
origin:         AS17660
notify:         netops@bt.bt
mnt-by:         MAINT-BT-DRUKNET
last-modified:  2010-07-21T03:46:02Z
source:         APNIC
```

IRR (Internet Routing Registry) lookup

- In some cases, inter-AS routing policies (aut-num objects)

```
aut-num:        AS17660
as-name:        DRUKNET-AS
descr:          DrukNet ISP
descr:          Bhutan Telecom
descr:          Thimphu
country:        BT
import:         from AS6461 action pref=100; accept ANY
export:         to AS6461 announce AS-DRUKNET-TRANSIT
import:         from AS2914 action pref=150; accept ANY
export:         to AS2914 announce AS-DRUKNET-TRANSIT
import:         from AS6453 action pref=100; accept ANY
export:         to AS6453 announce AS-DRUKNET-TRANSIT
```

IRR (Internet Routing Registry) lookup

- In some cases, transit provider policies (as-set objects)
  - Helps aggregate downstream networks!

```
route:        X.1.2.0/24
origin:       AS18024

route6:       2001:db8:a::/48
origin:       AS18024
```

```
route:        X.1.3.0/24
origin:       AS18025
```

```
route6:       2001:db8:b::/48
origin:       AS18025
```

```
➜ whois -h whois.apnic.net "AS-DRUKNET-TRANSIT"
% [whois.apnic.net]
% Whois data copyright terms    http://www.apnic

% Information related to 'AS-DRUKNET-TRANSIT'

as-set:         AS-DRUKNET-TRANSIT
descr:          DrukNet transit networks
members:        AS17660
members:        AS132232
members:        AS134715
members:        AS135666
members:        AS137925
members:        AS59219
members:        AS18024
members:        AS18025
members:        AS137994
members:        AS140695
members:        AS151498
members:        AS151955
members:        AS152317
members:        AS138558
members:        AS13335
members:        AS132894
members:        AS138529
members:        AS153779
members:        AS153740
admin-c:        DN01-AP
tech-c:         DN01-AP
notify:         netops@bt.bt
mnt-by:         MAINT-BT-DRUKNET
last-modified:  2025-06-09T11:07:14Z
source:         APNIC
```

Looking up as-set: PeeringDB

PeeringDB

Search here for a network, IX, or facility.

Advanced Search | Legacy Search

Registeror Login

English (English)

Bhutan Telecom Ltd

https://www.peeringdb.com

EXPORT

Organization	Bhutan Telecom Ltd
Also Known As	BT
Long Name	Bhutan Telecom Limited
Company Website	http://www.bt.bt
ASN	17660
IRR as-set/route-set	AS-DRUKNET-TRANSIT
Route Server URL	
Looking Glass URL	
Network Types	NSP
IPv4 Prefixes	100
IPv6 Prefixes	50
Traffic Levels	20-50Gbps
Traffic Ratios	Mostly Inbound
Geographic Scope	Not Disclosed
Protocols Supported	Unicast IPv4 Multicast IPv6 Never via route servers
Last Updated	2025-05-08T05:39:46Z
Public Peering Info Updated	2025-05-08T05:18:10Z
Peering Facility Info Updated	2025-05-08T05:38:11Z
Contact Info Updated	2025-05-08T05:39:39Z

Public Peering Exchange Points

Filter

Exchange / IPv4	ASN / IPv6	Speed / Port Location	RS Peer	BFD Support
DE-CIX Mumbai 103.27.171.100	17660 2401:7500:fff6::2e	10G	✓	○
Equinix Singapore 27.111.228.127	17660 2001:de8:4::1:7660:1	10G	○	○
Extreme IX Mumbai 103.77.108.239	17660 2001:df2:1900:2::239	10G	✓	○

Interconnection Facilities

Filter

Facility / ASN	Country / City
Bharti Airtel Mumbai 17660	India Mumbai
Equinix SG1 - Singapore 17660	Singapore Singapore

What is bgpq4?

https://github.com/bgp/bgpq4

- BGP filter generation tool → relies on IRR objects
  - prefix-lists
  - AS-PATH filters
  - Multi OS outputs
- Installation:

```
brew install bgpq4
```

```
git clone https://github.com/bgp/bgpq4
cd bgpq4
./bootstrap
./configure
make && sudo make install
```

Using route/route6 → prefix-list

- IPv4:
  - Without –S option

```
bgpq4 -l PEERv4-IN AS17660
bgpq4 -Jl PEERv4-IN AS17660
bgpq4 -Ul PEERv4-IN AS17660
bgpq4 -bl PEERv4-IN AS17660
```

  - With –S option

```
bgpq4 -S APNIC -l PEERv4-IN AS17660
bgpq4 -S APNIC -Jl PEERv4-IN AS17660
bgpq4 -S APNIC -Ul PEERv4-IN AS17660
bgpq4 -S APNIC -bl PEERv4-IN AS17660
```

- IPv6:
  - Without –S option

```
bgpq4 -6l PEERv6-IN AS17660
bgpq4 -6Jl PEERv6-IN AS17660
bgpq4 -6Ul PEERv6-IN AS17660
bgpq4 -6bl PEERv6-IN AS17660
```

  - With –S option

```
bgpq4 -S APNIC -6l PEERv6-IN AS17660
bgpq4 -S APNIC -6Jl PEERv6-IN AS17660
bgpq4 -S APNIC -6Ul PEERv6-IN AS17660
bgpq4 -S APNIC -6bl PEERv6-IN AS17660
```

Using as-set → as-path filter

- Without specifying source:

```
bgpq4 -l BT-AS-IN -f 17660 AS-DRUKNET-TRANSIT
bgpq4 -Jl BT-AS-IN -f 17660 AS-DRUKNET-TRANSIT
bgpq4 -Ul BT-AS-IN -f 17660 AS-DRUKNET-TRANSIT
bgpq4 -bl BT-AS-IN -f 17660 AS-DRUKNET-TRANSIT
```

- Specifying the source – Option1:

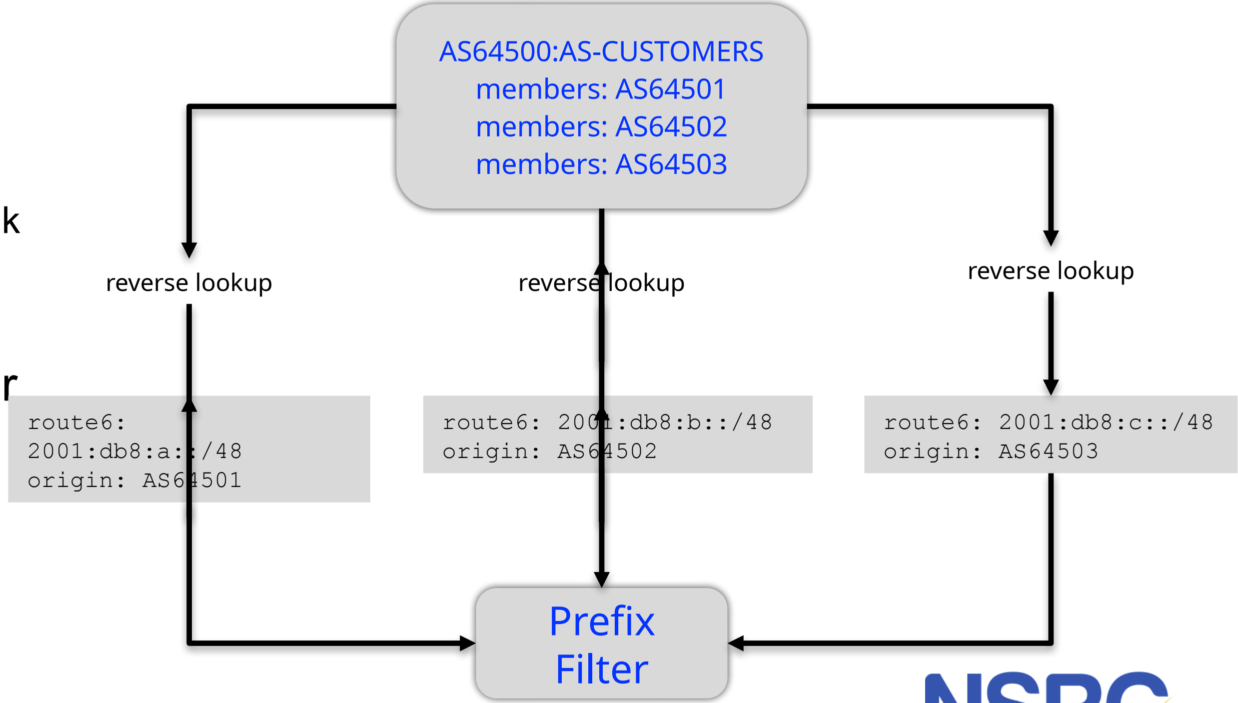
```
bgpq4 -S APNIC -l BT-AS-IN -f 17660 AS-DRUKNET-TRANSIT
bgpq4 -S APNIC -Jl BT-AS-IN -f 17660 AS-DRUKNET-TRANSIT
bgpq4 -S APNIC -Ul BT-AS-IN -f 17660 AS-DRUKNET-TRANSIT
bgpq4 -S APNIC -bl BT-AS-IN -f 17660 AS-DRUKNET-TRANSIT
```

- Specifying the source – Option2*:

```
bgpq4 -l BT-AS-IN -f 17660 APNIC::AS-DRUKNET-TRANSIT
bgpq4 -Jl BT-AS-IN -f 17660 APNIC::AS-DRUKNET-TRANSIT
bgpq4 -Ul BT-AS-IN -f 17660 APNIC::AS-DRUKNET-TRANSIT
bgpq4 -bl BT-AS-IN -f 17660 APNIC::AS-DRUKNET-TRANSIT
```

Using as-set → prefix-list

- Check the as-set
  - Walk the as-set and prepare a list of all the ASNs contained
  - If another as-set is contained, recursively walk it
- With the list of ASNs, run an inverse query for each one
  - Get the route objects where they are listed as origin



Using as-set → prefix-list

No source:

IPv4

```
bgpq4 -l PEERv4-IN AS-DRUKNET-TRANSIT
bgpq4 -Jl PEERv4-IN AS-DRUKNET-TRANSIT
bgpq4 -Ul PEERv4-IN AS-DRUKNET-TRANSIT
bgpq4 -bl PEERv4-IN AS-DRUKNET-TRANSIT
```

IPv6

```
bgpq4 -6l PEERv6-IN AS-DRUKNET-TRANSIT
bgpq4 -6Jl PEERv6-IN AS-DRUKNET-TRANSIT
bgpq4 -6Ul PEERv6-IN AS-DRUKNET-TRANSIT
bgpq4 -6bl PEERv6-IN AS-DRUKNET-TRANSIT
```

Specifying the source:

```
bgpq4 -S APNIC -l PEERv4-IN AS-DRUKNET-TRANSIT
bgpq4 -S APNIC -Jl PEERv4-IN AS-DRUKNET-TRANSIT
bgpq4 -S APNIC -Ul PEERv4-IN AS-DRUKNET-TRANSIT
bgpq4 -S APNIC -bl PEERv4-IN AS-DRUKNET-TRANSIT
```

```
bgpq4 -S APNIC -6l PEERv6-IN AS-DRUKNET-TRANSIT
bgpq4 -S APNIC -6Jl PEERv6-IN AS-DRUKNET-TRANSIT
bgpq4 -S APNIC -6Ul PEERv6-IN AS-DRUKNET-TRANSIT
bgpq4 -S APNIC -6Bl PEERv6-IN AS-DRUKNET-TRANSIT
```

AS-SET: Problem-1

- Be careful of bloated AS-SETs
  - https://www.kentik.com/blog/the-scourge-of-excessive-as-sets/
  - Eg: AS9498:AS-BHARTI-IN
    - https://bgp.tools/as-set/as9498:as-bharti-in
      - 68037 ASNs!
      - IPv6 prefix count: 807,572
      - IPv4 prefix count: 1,950,008

AS-SET: Problem-2

- Non-hierarchical as-set
  - What is wrong with AS-DRUKNET-TRANSIT?
  - Name collision!

```
as-set:         AS-AMAZON
descr:          Amazon ASNs
members:        AS-AMAZON-NA, AS-AMAZON-AP, AS-AMAZON-EU, AS16509:AS-AMAZON
admin-c:        AC6-ORG-ARIN
tech-c:         AC6-ORG-ARIN
notify:         noc@amazon.com
mnt-by:         MAINT-AS16509
changed:        noc@amazon.com 20230420  #17:54:10Z
source:         RADB

as-set:             AS-AMAZON
tech-c:             DUMY-RIPE
admin-c:            DUMY-RIPE
mnt-by:             KATERINA-MNT
created:            2022-10-23T19:05:59Z
last-modified:      2022-10-23T19:05:59Z
source:             RIPE
```

AS-SET: Problem-2

- Hierarchical as-set
  - RFC2622: <AS#>:AS-<as-set-name>

```
as-set:         AS4826:AS-VOCUS
descr:          Vocus Communications AS4826 AS-SET
members:        AS4826, AS4826:AS-CUSTOMERS
admin-c:        VPL1-AP
tech-c:         VPL1-AP
remarks:        For queries please email the below contacts
remarks:        NOC - noc@vocus.net
remarks:        IRR Data - irr@vocus.com.au
remarks:        Peering enquiries - peering@vocus.com.au
notify:         irr@vocus.com.au
mnt-by:         MAINT-AU-VOCUS
last-modified:  2022-05-29T00:28:23Z
source:         APNIC
```

```
as-set:         AS4826:AS-CUSTOMERS
descr:          Vocus Communications AS4826 Customer AS-SET
members:        AS10083,AS10146,AS10150,AS10204,AS10207,AS10213
members:        AS10234,AS11795,AS11857,AS12007,AS12041,AS12200
members:        AS12684,AS12740,AS131074,AS131162,AS131182,AS131214
members:        AS131282,AS131291,AS131298,AS131480,AS13150,AS131581
members:        AS131599,AS131715,AS131998,AS132030,AS132046,AS132094
members:        AS132145,AS132205,AS132221,AS132245,AS132257,AS132269
members:        AS132321,AS132368,AS132433,AS132448,AS132458,AS132476
members:        AS132481,AS132491,AS132492,AS132534,AS132576,AS132581
```

- Prop-151:
  - Restricts creation of non-hierarchical as-set
  - as-set can ONLY be created by the maintainer of the ASN in the object

AS-SET Transition Plan

- Adopted from Yoshinobu Matsuzaki's talk:
  - Step-1: Create a new hierarchical as-set
    - Eg: `AS17660:AS-BT`
    - Make sure - same members in the new as-set as your current as-set
  - Step-2: Add the new as-set (`AS17660:AS-BT`) as a member of the old as-set (`AS-DRUKNET-TRANSIT`)
    - From here on, you only need to update your `AS17660:AS-BT`
  - Step-3: Talk to your peers
    - Update **PeeringDB** entry
    - Ask your peers/upstreams to use your new as-set
  - Step-4: Then delete the old as-set!

AS-SET Transition Plan

Example: GEMNET - Mongolia

```
as-set:         AS-GEMNET
descr:          GEMNET LLC
country:        MN
members:        AS9934, AS9484, AS10219, AS9789,
                AS38038, AS24496, AS24559, AS4850,
                AS38042, AS38051, AS10076,
                AS17882, AS38218, AS38454, AS38805,
                AS38818, AS38273, AS10109, AS38869,
                AS45237, AS45509, AS17463, AS24302,
                AS45203, AS45204, AS45923, AS24320,
                AS24496, AS45237, AS55342, AS55805,
                AS55801, AS55408, AS55882, AS56194,
                AS58598, AS58625, AS133177, AS135136, AS135033, AS131240, AS136437, AS141681, AS136936,
                AS138907, AS134074, AS45204:AS-GEMNET
tech-c:         GA263-AP
admin-c:        GA263-AP
mnt-by:         MAINT-GEMNET-MN
mnt-lower:      MAINT-GEMNET-MN
last-modified:  2024-10-08T02:49:37Z
source:         APNIC
```

```
as-set:         AS45204:AS-GEMNET
descr:          as set for GEMNET LLC
country:        MN
members:        AS9934,AS9484,AS10219,AS9789,AS38038,AS56301
members:        AS24496,AS24559,AS4850,AS38042,AS38051
members:        AS10076,AS17882,AS38218,AS38454,AS38805,AS38818
members:        AS38273,AS10109,AS38869,AS45237,AS45509,AS17463
members:        AS24302,AS45203,AS45204,AS131199,AS45923,AS55342
members:        AS24320,AS55805,AS55801,AS55408,AS55882,AS56194
members:        AS56293,AS58439,AS58598,AS58625,AS133177,AS58726
members:        AS58406,AS59354,AS133453,AS63962,AS134074,AS134241
members:        AS135033,AS135136,AS131240,AS18172,AS136254,AS136936
members:        AS137261,AS137511,AS56038,AS13335,AS137982,AS133050,AS138907
members:        AS139579,AS134356,AS141681,AS152337,AS152692,AS151637,AS152299,AS139057
tech-c:         GA263-AP
admin-c:        GA263-AP
mnt-lower:      MAINT-GEMNET-MN
mnt-by:         MAINT-GEMNET-MN
last-modified:  2025-04-16T03:41:33Z
source:         APNIC
```

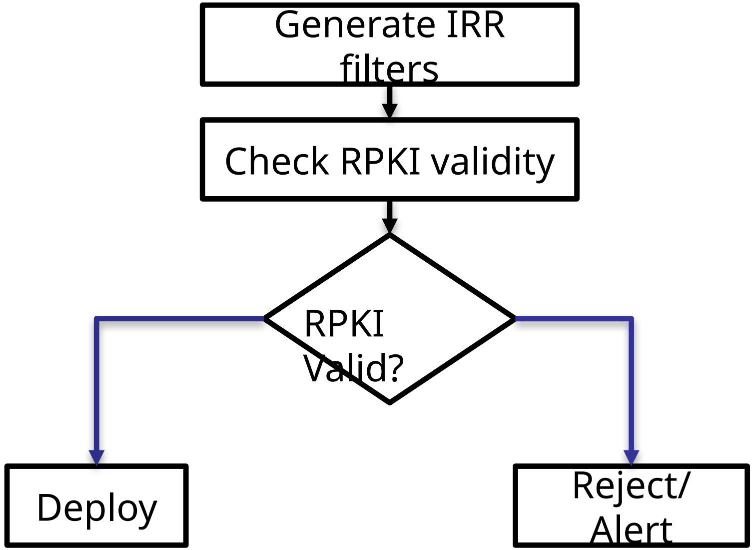
Organization	Gemnet LLC
Also Known As	
Long Name	
Company Website	https://www.gemnet.mn
ASN	45204
IRR as-set/route-set	AS-GEMNET

Problem with IRR

- Incomplete or stale data
  - Logic: accept what's in the IRR and reject the rest!
    - If a `route(6)` is not in IRR – is it invalid or the IRR database missing data?
    - No automatic cleanup – networks don't remove IRR entries?
    - Manual updates required – people forget to update `as-set/route(6)` objects
- No single authority model – fragmented databases
  - How do you know if an entry is genuine?
  - If two IRRs contain conflicting date, which one to trust?
    - Use authoritative sources: -S option (APNIC; RIPE, etc)

Pair IRR with RPKI

- Once you have generated IRR based filters
  - Check against RPKI for validity: `Origin AS, Prefix, Prefix Length`



Questions?

UNIVERSITY OF OREGON

NSRC
Network Startup Resource Center